How the Life Sciences are Benefiting from Open Source Software

This article gives an overview of the role of open source software and computer technology in the field of life sciences, and how research and discovery in the fields of medicine, biology, etc, is enhanced by the use of both.

The primary quest of man (barring survival and self-preservation) has been to understand his place in the universe. What distinguishes planet Earth from the rest of the known universe is the presence of life. Arguably, the fundamental problems faced by humans are life science problems.

Computational sciences and the life sciences were earlier treated as totally different and contrasting fields. The emergence of bioinformatics shattered that mindset. However, with the emergence of newer challenges, interdisciplinary approaches to studying the life sciences are becoming more popular. Relatively new computational approaches to the life sciences are now showing a lot of promise, and opening up exciting new frontiers in systems and computational biology. Computational science is being used not just to analyse data but to model and simulate living organisms.

Open source software and technology are heavily exploited in these efforts, and it would not be a stretch to say that this keeps the research open and freely accessible.

The data explosion

Today, data produced by experiments in life sciences is increasing at a rate that is twice that of the storage capacity of computer hardware. The rate of the availability of new data has already outpaced our ability to make sense of it. Indeed, life sciences data today is accessed as much by non-life sciences professionals as by those in the life sciences field.

The genomes are one of the main causes of the data explosion. A genome is the encoding of an organism's DNA, so to speak, and is a reflection of the complexity of life. The 'code size' of the genome (measured in number of base pairs) ranges from 1.8Kbp (Kilo base pairs) for a virus to about 3.2Gbp for humans. While the human genome is less than a gigabyte, it is customary to store 100 times as much data per individual to eliminate errors. With the cost of DNA sequencing falling by orders of magnitude over the past few years, the storage requirements for the genomes of a sizeable population will certainly strain the available storage capacity on the planet. Clearly, newer approaches to minimising storage requirements are needed and are being actively explored.

The Ensembl (*ensembl.org*) genome browser-cum-database uses open source technology (MySQL, Perl, Apache, Git, etc) not just to make the genomic and other data interactively accessible, but also to enable researchers to annotate and update genomic data. It is a massive collaborative effort. In true open source spirit, it has a developer site and allows you to install and host your own Ensembl site.

A genome database is quite unlike other databases, for instance, those that have details about cell phone users or registered voters. This is because genomic data is continuously changing. The changes could be because the data becomes more species-specific and more complex. As more research is conducted using the model organisms whose genome databases are available, more details get added.

While existing database technology is already in use in most cases, the need for newer paradigms is being felt, especially to help explore the complex interrelationships of the data and to search for complex sequences. Technologists are on the lookout for an approach to integrate various types of biological data, which would make it easier to draw more useful conclusions. Bio4j (*bio4j.com*) is a graph database, which is also a bioinformatics platform integrating data from